Semiconductor Monthly

Discrete Semiconductors

Improving Power Factor Correction & Reducing EMI in Switch-Mode Power Supply Design

by Ronald H. Randall, Fairchild Semiconductor

Switch-mode power supplies (SMPS) are increasingly being designed with an active power factor correction (PFC) input stage to meet international regulatory standards for harmonic content. Historically, power factor correction circuits have used a boost converter topology that combined a power switch and boost diode. Due to the recovery losses of the diode, snubber circuitry was necessary to reduce turn-on losses and electro-magnetic interference (EMI). However, with the recent introduction of the soft recovery hyperfast "stealth" diode, snubber circuitry is no longer necessary, and the boost converter can be implemented in the hard-switched mode. When the new diode is combined with advances in insulated gate bipolar transistor (IGBT) technology, designers can obtain lower conduction losses, easier gate drive and better Vsat/Eoff trade offs. In addition, previous limitations to PFC design may be overcome and optimal PFC performance achieved, making it easier to meet harmonic standards such as the EN 61000-3-2.

In traditional switching power supplies the AC source is rectified into a large capacitor filter and current is drawn from the source in narrow high amplitude pulses. The high amplitude current pulses produce harmonics that can cause severe interference with other equipment and reduce the maximum power that can be drawn. Utilizing power factor correction not only ensures compliance with regulatory specifications, but also improves device efficiency by increasing the maximum power that can be drawn from the source.

Instituting laws and restrictions that require active power factor control has been an on-going effort since 1992, with the most recent approved standard being EN 61000-3-2 with Amendment A14. The objective of this standard is to limit the magnitude of the individual harmonic currents up to the 39th harmonic. While Amendment A14 is targeted at relaxing the requirements for some equipment currently in the "Class D" grouping, equipment such as personal computers, monitors and television sets are still maintained under the original EN 61000-3-2 "Class D" requirements.

Power Factor Correction Basics

Power factor correction refers to the process by which the input is made to look more like a resistor. This is desirable since the resistor has a unity power factor, compared to the typical SMPS power factor values of 0.6 to 0.7, and allows the power distribution system to operate at its maximum efficiency. Power factor is defined as the ratio of real power to apparent power. Real power is the power measured by a wattmeter

$$Watts = \frac{1}{T} \cdot \int_0^T I(t) \cdot V(t) \cdot dt$$

and apparent power is the product of rms current times rms voltage. The rms current is

$$Irms = \sqrt{I_1^2 + I_2^2 \cdots I_n^2}$$

where n is the order of the respective harmonic currents (i.e., 2nd, 3rd, etc.). Making the power supply input current approach a sinusoid removes the harmonic currents, decreasing the rms input current and increasing the circuit power factor. The result is maximum utilization of the source power and reduction of harmonic content.

Most active PFC designs incorporate the continuous current mode (CCM) boost converter topology because of its simplicity and broad AC input voltage range. The boost converter, shown in Figure 1, places the boost diode D5 and switching device S1 in the hard-switched mode. The drawback to hard switching is that the diode reverse recovery characteristics increase the switching device's turn-on losses and the generated EMI.

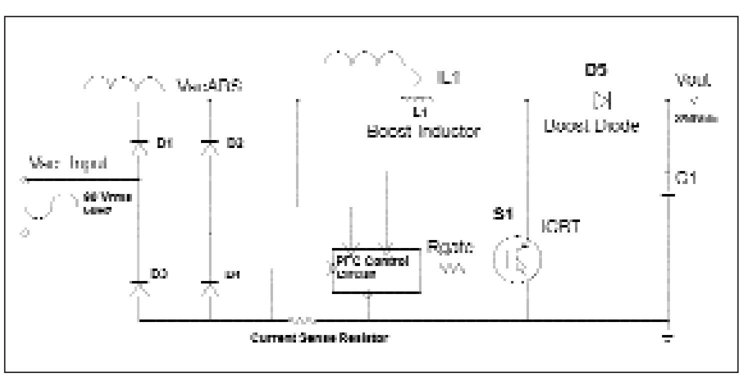

Figure 1 Boost PFC Circuit

The diode's reverse recovery characteristics describe how the device transitions from the forward conducting state to the reverse voltage blocking state. If the return of the reverse recovery current from I_{RRM} to zero is too abrupt or "snappy," voltage spikes and severe EMI are generated. To soften this response, designers have either slowed down the switch turn on di/dt and/or added snubber circuits. With prior diode technology, the designer was limited to having either a soft or fast diode. The large I_{RRM} value of previous diode technology generates large switching device turn-on loss during the diode t_{rr} recovery period.

Slowing down the switch turn-on rate increases the switch turn-on loss. Adding snubber circuitry adds to circuit cost and complexity and reduces circuit reliability. The snubber circuits often involve complex energy recovery schemes since the basic RC approach results in high power dissipation in the snubber resistor.

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RapidIO™ Spec Released

by Carol Rosen, Western Regional Editor

Most often in the world of electronics, IC designers follow system specifications in developing ICs to eliminate problems. Often, these specs are developed to solve current and future generation problems. Recently, the nearly 1-year-old RapidIO™ Trade Assn. offered a free, public release of its RapidIO interconnect specification.

This spec is designed to speed and improve communication among chips or boards within a system. It defines a switch fabric-based control plane interconnect offering a dramatic increase in bandwidth in addition to more scalability and higher reliability than bus-based control plane interconnects. Most important, it's compatible with existing architectures.

The specification defines a switch fabric architecture. It's backed by a number of industry vendors including Alcatel, Cisco Systems, EMC Corp., Ericsson, Lucent Technologies, Mercury Computer Systems, Motorola and Nortel Networks. The spec provides transmission rates hundreds of times faster than current interconnect architectures. It also inter-operates with PCIBus and is software transparent.

Current chip-to-chip and board-to-board communication is limited to transmission rates of hundreds of megabits per second in many systems. RapidIO offers performance levels up to 64 Gbps, per port, with an aggregate of hundreds of gigabits per second in a single system. It can be used in a broad range of applications including networking and telecommunication equipment, storage subsystems, signal and imaging processing and other embedded computing and communication systems. The specification can be integrated into these applications quickly and easily since FPGA chips can be programmed to support RapidIO.

The specification itself is a conglomerate of real-time performance with scalability and flexibility. It contains hardware-based error detection and correction on each link in the data transfer path with guaranteed forward progress. The technology offers a small silicon footprint with a low pin count.

The specification architecture is divided into three hierarchical segments: logical, transport and physical layers. Each layer can be modified or new layers added without disturbing other elements in the technology. One example, is the addition of a serial communication physical layer, which is the focus of the serial working group within the RapidIO Trade Assn.

Developed through collaborative efforts among the associations members representing networking, telecommunications and manufacturing, RapidIO Interconnect Specification 1.1 can be downloaded at http://www.rapidio.org/specs.htm for no cost.

Intarsia Develops Amplifier Modules

This month Intarsia Corp. is announcing seven new amplifier modules. The first four are part of a family of low noise amplifier modules whose frequencies range from 800 MHz to 6.0 GHz. The other three are power amplifier modules for wireless and RF systems.

The low noise amplifiers are designed for broadband wireless access equipment, cellular base stations and infrastructure equipment for emerging wireless LAN, 802.11a, 802.11b and Bluetooth systems. They provide noise figure performance down to 0.65 dB and are developed as drop in solutions to eliminate external components. The new modules use Intarsia's thin-film on-glass technology to incorporate complete matching networks and all bias and de-coupling components onto a single glass substrate. The company packages this component with active RF chips in industry-standard plastic housings.

The new modules contain matched inputs and outputs to 50 Ω without external biasing and de-coupling. By incorporating supporting passive networks, Intarsia eliminates yield issues due to tolerances of various individual components. A single module increases circuit board level yields and saves board space. Additionally, the modules reduce RF losses because the integrated passive network is connected by wirebonds in the same package to the RF chip. This lower noise allows higher levels of receiver sensitivity providing

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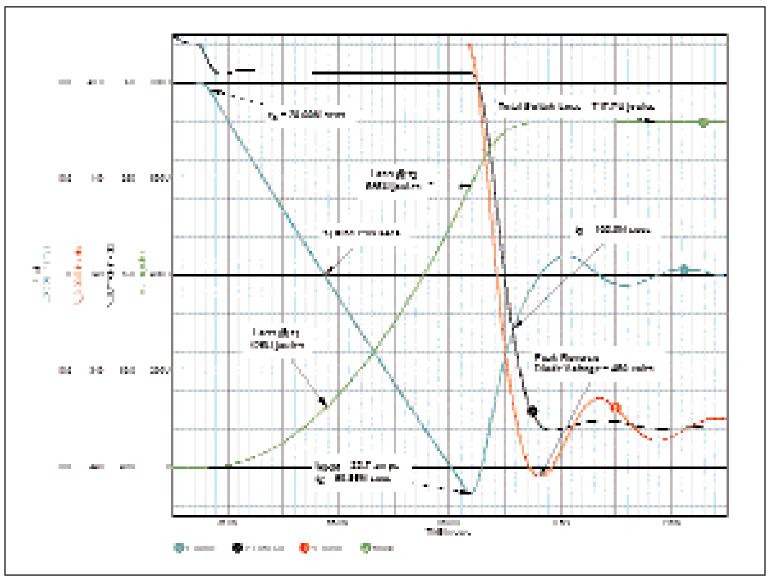

Figure 2 Hard Switched Turn-On Recovery of "Snappy" Diode

Hard-switched turn-on tradeoffs are better understood by breaking the switch turn-on period into discrete time segments. Figure 2 illustrates the recovery of a snappy diode in the circuit. The diode is conducting 20 A and commutated by a switch turning on with a di/dt of 600 A/µs. The circuit model for Figure 2 contains 50 nH loop inductance in the hard-switch recovery path of C1, D5 and S1 to represent the stray wiring circuit inductance. A 24 Ω resistor and 0.20 nF capacitor RC snubber were required across the boost diode to keep the diode peak reverse voltage within its maximum rating. Losses in the snubber resistor are 18 µJ representing 1.8 W loss at 100 kHz. The total snubber loss is twice this value, 3.6 W, since the energy in the capacitor is dissipated in the snubber resistor when the switch is turned off. The green waveform in Figure 2 is the integral of the product of the switch voltage and the switch current. This trace represents the energy lost in the switch versus time.

During the first period, t_1, the switching device current increases to the 20 A diode forward-conduction current value and the diode current reduces to zero. The potential across the switch is nearly 400 V DC during t_1. In the example, 125 µJ or 17.4 percent of the total switch turn-on loss occurs during t_1. This time period and the losses associated with it are a function of the switch turn-on di/dt rate. Unfortunately, slowing down the switch turn-on di/dt to minimize EMI and reduce the diode reverse voltage spike increases the t_1 time and its associated losses.

The second period, $t_a = t_2 - t_1$, when the diode current reverses to its I_{RRM} value, is both diode and power circuit dependent. The diode I_{RRM} is a function of the recovery current di/dt and the diode junction temperature. Increasing either di/dt or junction temperature will increase the IRRM. The major portion of the switch turn-on loss occurs during t_a. In Figure 2, the turn-on loss during t_a is 460 µJ.

During the third period, t_b, the diode recovery period, the voltage across the switch reduces to its saturated value and the diode blocks the output voltage. In Figure 2, t_b is very short (11.6 ns) with a loss of 132.7 µJ in the switch. In most circuit designs, the inherent switch voltage fall time extends this period. Typical designs contain MOSFETs as the switching device and the MOSFET input capacitance, CISS, and reverse transfer capacitance, CRSS, increase the switch voltage fall time, resulting in additional switch losses. The total circuit loss during t_b is a function of the diode current since this current exists at the full 400 V output potential. The rate of switch voltage fall during t_b determines the distribution of this loss between the diode and the switch.

New Components Improve PFC Design

Recent developments in IGBT and diode technology provide a greatly improved component pair for the hard-switched boost PFC topology. Advances in IGBT switching performance, as exhibited by the HGTG12N60A4 in Fairchild's SMPS IGBT series, provide a device capable of hard-switched operation up to 100 kHz. These IGBTs have a much lower on-state voltage when compared to MOSFETs having equivalent die size (i.e., 1.7 V typical, 2 V max for the HGTG12N60A4 IGBT versus 9.5 V for an IRFP450 MOSFET @ 12 A and 125°C. They also have significantly lower input and reverse transfer capacitances, typically C_{IES}=2.5 nfd and C_{RES}=1.5 nfd @ 1 V for the HGTP12N60A4 IGBT, compared to C_{ISS}=5.5 nfd and C_{RSS}=3.5 nfd @ 1 V for the IRFP450. These values produce a much shorter voltage fall time and have half of the turn-off losses of previous generations. The combination of these IGBTs with new hyperfast stealth diodes is a practical approach to low-cost, reliable PFC circuits. The new diodes are rated at 600 V and combine fast recovery characteristics of hyperfast technology with soft recovery characteristics to achieve lower switching losses and low EMI.

Table 1 Boost switch and diode pairs for various PFC power levels and input voltage ranges.

Minimum AC voltage	PFC Output Power(W)	Stealth Diode	IGBT
80	175	ISL9R460P2	HGTG7N60A4 or
180	400		HGTG12N60A4
80	400	ISL9R860P2	HGTG7N60A4 or
180	850		HGTG12N60A4
80	750	ISL9R1560P2	HGTG12N60A4 or
180	1700		HGTG20N60A4
80	1500	ISL9R3060G2	HGTG30N60A4 or
180	3400		HGTG40N60A4

They are designed for use in high power, high frequency circuits and are ideally suited for PFC circuits as a boost diode, SMPS, UPS, and motor drives to name a few typical applications. A diode's softness rating, S, is defined as t_b/t_a. Typical values for commonly available diodes range from 0.3 to 0.6. The new diodes provide softness values greater than 1.2 at a temperature of 125°C and have a t_{rr} reverse recovery time of less than 25 ns.

Figure 3 illustrates the advantages of a soft recovery diode in the PFC circuit of circuit. The conditions of Figure 3 differ from Figure 2 in that the diode RC snubber is removed and the snappy diode is replaced with a soft-recovery device. The stealth technology provides a diode that recovers in a soft monotonic mode, resulting in lower EMI. In addition, the ratio of I_{RRM}/I_{fwd} is decreased, resulting in lower t_a losses. The lower I_{RRM} allows the designer to turn on the switch at a higher di/dt rate (in some cases over 600 A/µs), reducing the t_1 losses in the IGBT by as much as 30 percent. The switch t_1, t_a and t_b turn-on losses are 121, 259 and 149 µJ, respectively. The soft recovery diode not only results in a reduction in the switch losses but it also permits the removal of the parallel RC snubber circuit.

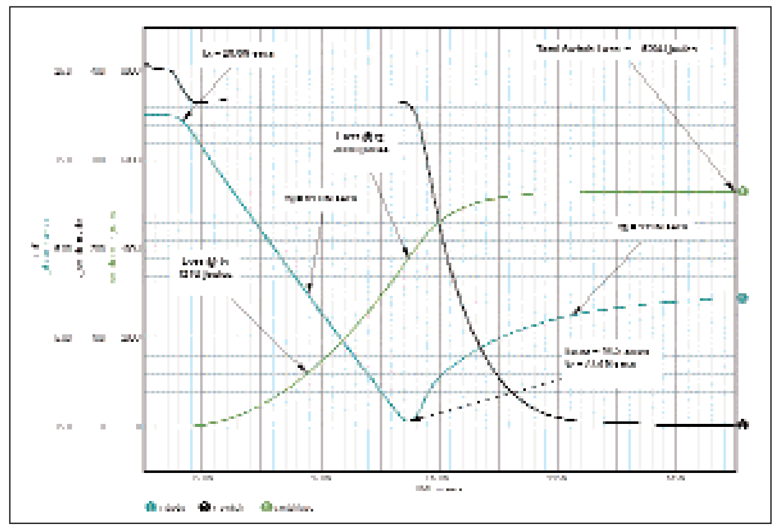

Figure 3 Hard-Switched Turn-On Recovery of Soft or Stealth Diode

Table 1 gives the suggested stealth diode and IGBT combinations as a function of PFC circuit power level and AC input. These are only recommendations since the IGBT and stealth diode selection will also depend on heat sink size and its thermal performance. (*The information provided in this table was compiled through the application efforts of Sampat Shekhawat, Fairchild Semiconductor Staff Engineer*).

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2. Sampat Shekhawat, et al. "Intersil's New Stealth Diode Reduces SMPS IGBT Switching Loss," PCIM-2000, Boston, MA.

3. Sampat Shekhawat, et al. "New Stealth Soft Recovery Diode Reduces SMPS IGBT Switching Loss," PCIM-2000, June 19-21, 2000, Nuremberg, Germany.

Ronald H. Randall received a BS in Electrical Engineering from Tri-State University, Angola, Indiana in 1965. He has been a member of IEEE and worked in power electronics design and development for 35 years. Ron holds five patents in the power electronics field. His recent activity has been in the characterization and application of IGBTs in SMPS applications. His main interests are in the development of simulation and modeling techniques for power electronic circuits.

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increased range and cell size.

The four chips include the LNA-015-01-S08, a 1.5 GHz to 2.7 GHz low noise amplifier with a minimum noise figure of 0.65 dB. Its target applications include broadband wireless access equipment, cellular base stations and repeaters, MMDS, 2.45 GHz W-LAN and Bluetooth systems. The LNA-020-01-S08 achieves 2.0 GHz to 2.75 GHz with a minimum noise figure of 0.75 dB. It is designed for BWA and MMDS at 2.5 GHz, 802.11b infrastructure and emerging Bluetooth access points. The LNA-030-01-S08 offers a 3.0 GHz to 6.0 GHz range and a minimum noise figure of 0.9 dB. It's targeted at BWA, MMDS, wireless local loop and 5 GHz 802.11a W-LAN and HiperLAN infrastructures. It also can be used as an intermediate frequency amplifier for local multi-point distribution services and applications in the 20 GHz to 30 GHz range. The LNA-088-03-S08 is a single-band LNA device for cellular equipment operating in the 800 MHz to 900 MHz range. It offers a minimum guaranteed 1.2 dB noise figure.

The four will sample beginning May 18. Production quantities are scheduled for September. The four LNA's prices range from $2 to $3 each in 10,000-piece quantities.

The three power amplifier modules being announced May 18 are developed for single-band or dual-band cellular phones and base stations in the PCS frequency band between 1.7 Ghz and 1.9 GHz. They also use the company's thin-film on-glass technology. Additional power and driver amplifier modules will be released in the next six to nine months.

The technology also provides time savings since system designers don't have to develop matching networks with numerous discrete devices, yield issues are eliminated and board space is saved. The first three-power amplifier modules — PAM-018-03-M48, PAM-018-04-M48 and PAM-018-050-M48 are three stage amplifiers with inputs and outputs matched to 50 Ω. They use PHEMT technology and initial offers are in 7 mm × 7 mm × 1.2 mm plastic packages.

The PAM-018-03-M48 is a broadband power amplifier operating between 1.75 GHz and 1.91 GHz, making it suitable for Korean and US PCS systems. The PAM-018-04-M48 as a narrow band PA that operates between 1.75 GHz and 1.78 GHz, and was designed especially for Korean CDMA PCS uses. The PAM-018-05-M48 offers 1.85 GHz to 1.81 GHz frequencies and is aimed at US CMDA and TDMA PCS markets.

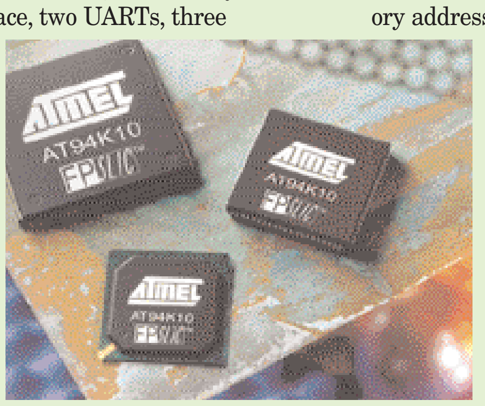
Currently sampling, all three are scheduled for volume production in September. They are priced under $5 each in quantities of 10,000 pieces. **Intarsia Corp. , 48611 Warm Springs Blvd., Fremont, CA 94539; (510) 354-6301; Fax: (510) 354-6330; www.intarsiacorp.com.**

Write in 5222 or **www.ecnmag.com/info**

Atmel Unveils Programmable SoC

Atmel Corp. is introducing a programmable SoC that combines an MCU, programmable logic, control, memory and I/O functions, but consumes 2 mA to 3 mA per MHz during operation and under 50 µA in standby. It is targeted at low power applications including PDAs and their peripherals, cellular phone add-on equipment GPS, portable test equipment, point-of-sale equipment, security systems and wireless Internet applications.

The AT94L10 is the second member of the company's RISC-based programmable SoC family. The new device incorporates a 20 plus MIPS AVR RISC microcontrollers with a hardware multiplier, 10,000 gates of SRAM-based programmable logic, 36 KB of SRAM, an industry standard two-wire serial interface, two UARTs, three timer/counters and a watchdog timer. The MCU contains 127 16-bit fixed length instructions optimized to enable 90 percent of MCU operations to be handled by one instruction. Its two-stage pipeline and separate addresses for load and store operations allow single-cycle instruction execution. It has 32, 8-bit registers permitting the CPU to access multiple data simultaneously.



Atmel Corp.'s new AT94K10 is a filed programmable system level IC combining an MCU, programmable logic, control, memory and I/O functions. It consumes under 50 µA in standby.

Atmel offers extensive EDA support with a co-verification framework permitting the MCU code to be debugged simultaneously with the HDL simulation of the FPGA logic. The core can be reconfigured in-system at the cell levels allowing designers to implement multiple different system functions. Available now in production quantities, the AT94K10 is housed in 84-pin PLCCs, 100-pin and 144-pin TQFPs and 208-pin PQFPs. Prices begin at $9.95 in 250,000-piece quantities. **Atmel Corp., 2325 Orchard Prkwy., San Jose, CA 95131; (408) 441-0311; Fax: (408) 487-2600; www.atmel.com.**

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PLX Introduces Controller, DSP Module

PLX Technology recently announced the GigaBridge™ GBP32 controller and the PLX POM62 DSP module. It developed the former to increase scalability and boost high-availability in PCI-based telecommunications, data communications and embedded systems. The module is a DSP engine designed to accelerate development of communication system designs based on the PLX IOP 480 I/O processor and TI's DSP technology.

PLX Technology's new GigaBridge™ GBP32 controller is based on the company's adaptive switch fabric architecture enabling implementation of PCI hard- and software-transparent switch fabrics.

The GBP32 controller is based on PLX's adaptive switch fabric architecture, which is designed to preserve existing PCI hardware, software and infrastructure compatibility. The adaptive switch fabric is a cell-based fabric with independent PCI bus segments connected to each port. Each GBP32 controller can drive up to four PCI slots and inter-operates with other GBP32 controllers as ports on the fabric.

The controllers are linked by two 16-bit wide, point-to-point, low-voltage differential links and clocked at 400 MHz. This offers an aggregate fabric bandwidth of 51.2 Gbps. The chip's cell processing engine manages how cells are routed from source to destination. It also automatically segments PCI transactions into fabric cell transactions and reassembles them back into PCI transactions.

The GigaBridge architecture permits PCI-based telecommunications, data communications and embedded systems to incorporate 224 bus segments, aggregate tens of gigabits per second and direct connection of two adjacent PCI devices on the fabric up to 15 feet away through a high-performance link.

Multiple GBP32s can inter-operate to form adaptive switch fabric, a 100 per cent PCI-compatible virtual backplane with up to 224 bus segments and up to 896 PCI slots. Within an eight port adaptive switch fabric configuration, up to eight concurrent transactions can occur, for 51.2 Gbps of aggregate bandwidth. Each GBP32 controls a PCI bus segment and permits any PCI device to link to any other PCI device on the fabric, up to a 15-foot difference.

The controller integrates flow control, automatic memory address space management, self-healing and bandwidth management. These enable fault isolation and containment down to a field replacement unit. CompactPCI hot swap and live insertion are supported at the board, fabric, box and rack level. A 64-bit GigaBridge controller is scheduled for introduction later this year. Development tools from PLX and third-party vendors will be shipping in the third quarter.

Available now, the GigaBridge GBP32 is housed in a 388-pin PBGA. Prices start at $54 each in quantities of 5,000-units. \

The POM62 DSP features a 200 MHz TI TMS320C6202 DSP, 4 MB of SDRAM and 512 KB of flash, a single interface between the DSP an IOP and an array of development software.

The module expands IOP 480's design options. When used in conjunction with PLX's IOP 480RDK, it offers a development system for TMS320C6202-IOP 480 combination so designers don't have to build their own hardware platform. It offers developers an easy method of developing cards and systems that take advantage of IOP 480's ability to offload communications processing tasks from the TMS230C6202.

IOP 480 connects to the DSP's Xbus offering a secondary bus for unimpeded transfers between the DSP memory, IOP 480 local memory and the PCI bus. The POM62 module plugs into the PLX option module connector on the IOP 480RDK board freeing the DSP's external memory interface bus from PCI-related activity and ensuring maximum DSP performance in compute intensive functions. The module includes test software compatible with TI's Code Computer Studio™ IDE and the DSP/bios real-time kernal.

Available in the third quarter, the PLX POM62 DSP module is priced at $495. **PLX Technology Inc., 870 Maude Ave., Sunnyvale, CA 94085; (800) 759-3735; Fax: (408) 774-2169; www.plxtech.com.**

Write in 5224 or **www.ecnmag.com/info**

VAutomation Offers USB 2.0 Controller Core, TotalCore

VAutomation, Inc. recently unveiled a new USB 2.0 device controller core. The VUSB-HS is a synthesizable USB device controller core allowing a single piece of hardware to serve the different needs of many products and to adapt to evolving specifications.

The cores will connect to a 32-bit basic VCI-complaint system bus interface and 16-bit/8-bit USB transceiver macrocell interface. An integrated DMA controller alleviates the application processor of routine data movement chores. Small, internal latency buffers permit the VUSB-HS to be implemented in various types of hardware from 30 MHz FPGAs to 20 MHz ASICs.

The core is designed, except for one state of UTMI logic, in a single clock domain to ease synthesis, timing and verification of the supplied VHDL or Verilog source code. USB endpoint data are stored in system memory allowing designers to control the efficient use of the resource. The core complies with the Enhanced Host Controller Interface. Available this month, the core's price is based on a standard pricing sheet with licensing fees and royalties.

The company also introduced the iCon186 TotalCore. This bundled collected of IP combines VAutomation's high performance Turbo186 MPU with connectivity peripheral cores for USB, Ethernet and CAN into a synthesizable SoC platform.

iCon186 targets the 16-bit market so designers concerned with performance, code density and preservation of software can preserve their investments. The company redesigned the Turbo186 to achieve three times the performance of off-the-shelf 80186 devices while preserving legacy code investments with 100 per cent software compatibility.

The MPU provides selectable 20-/24-bit addressing. It also integrates the company's firmware-based dual-purpose VUSB 1.1 host/device, fast Ethernet MAC and CAN 2.0b controller for a stable, flexible platform that can be customized. Fully synchronous, static VHDL or Verilog is suitable for any FPGA, ASIC or foundry technology.

Available this month, the iCon186's price is dependent on various factors but follows the company's licensing free and royalty schedule. **Vautomoation, Inc./ARC Cores , 6204 San Ignacio Ave., San Jose, CA 95119; (408) 361-7800; www.VAutomation.com.**

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I-Cube Unveils New Crosspoint Switch

I-Cube, Inc. recently introduced the MSX340. The newest member of its MSX™ digital crosspoint switch family, the chip is SRAM-based and in-system programmable. It is designed for telecommunications systems including digital cross connects and other access products.

MSX340 is designed using the company's non-blocking crossbar architecture. This uses the switch matrix to connect input/output buffers in one-to-one or one-to-many connections. It features a non-blocking switch matrix with 340 individual configurable I/O ports and double buffered configuration RAM cells. It offers all the high performance features of the MSX532 with fewer ports, and offers an aggregate bandwidth of more than 25 Gbps for bit-oriented

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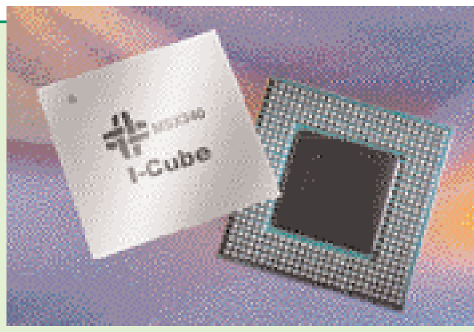
uses.

The chip contains individually configured I/O buffers that can be programmed to serve as inputs or outputs, to provide bi-directional operation or to operate in bus repeater™ mode. It also permits designers to select signal direction and data flow mode.



I-Cube's new crosspoint switch is designed in the company's non-blocking crossbar architecture that uses the switch matrix to connect input/output buffers in one-to-one or one-to-many connections.

Other features include next-neighbor clocking, which permits any I/O pin to be used as a clock source for an adjacent I/O pin and lets the IC independently switch different speed inputs at the same time. The MSX340 's double buffered configurable SRAM cells allow the chip to be used for bit switching and bus switching by enabling or disabling the feature. The chip also provides tight skew and identical pin-to-pin delays, a flow-through non-return to zero data rate of 150 Mbps and registered clock frequencies of 75 MHz. It operates at 3.3 V with LVTTL I/Os.

System designers configure the chip's switch matrix, I/O buffers and control registers with a JTAG serial interface or RapidConfigure™ interface. The later parallel interface has 28 pins. The JTAG interface is complaint with the IEEE-1149.1 with five pins for boundary scan testing in addition to device configuration and verification.

Housed in a 480-BGA, the MSX340 is available in volume now. It's priced at $134 each in 1,000-piece quantities. **I-Cube, Inc., 2605 S. Winchester Blvd., Campbell, CA 95008; (408) 341-1888; Fax: (408) 341-1899; www.icube.com.**

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LSI Logic Develops Integrated Mirroring

LSI Logic Corp. recently announced integrated mirroring. Part of the company's Fusion-MPT architecture, integrated mirroring offers data protection for the system boot volume to eliminate loss of critical information including operating systems on servers and high performance workstations.

The feature allows seamless and transparent fault tolerance, eliminating requirements for complex backup software or RAID hardware. It operates independently from the operating system.

It allows high level I/O performance with a message passing interface for value-added applications. The architecture is based on ARM processor technology. Its unique feature of a single binary device driver supporting Ultra 320, SCSI and Fibre Channel eliminates requirements for developing, installing and qualifying separate device drivers for each physical interface.

LSI Logic so far has developed five standard ICs in Fusion-MPT architecture. These include three Fibre Channel controllers; LSIFC929 PCI to dual channel 2 Gb, LSIFC919 PCI to single channel 2 Gb and the LSIFC909 PCI to single channel 1 Gb. There are two Ultra320 SCSI controllers, the LSI53C1030 PCI-X to dual channel and the LSI53C1020 PCI-X to single channel. The LSIFC909 is available now with the other two Fibre Channel devices scheduled for volume next month. The two Ultra320 SCSI are available this month. In single quantities, the LSI53C1030 is priced at $154 and the LSI53C929 at $172. Other prices will be announced later. **LSI Logic Corp., 1551 McCarthy Blvd., Milpitas, CA 95035; (866) 574-5741;www.lsilogic.com.**

Write in 5227 or **www.ecnmag.com/info**

Equator Introduces New Media Library

Equator Technologies, Inc. recently announced an enhanced media library with four new cores. The new devices include an MPEG2 decoder, MPEG4 decoder, H.263 encoder and H.263 decoder software. Available in source code form, the library offers a video technology base for the company's MAP-CA™ broadband signal processor platform.

The cores are designed for a variety of applications including set-top boxes, DSL and cable modems, video-on-demand platforms and traditional media companies. They were developed to provide system designers with the ability to develop new services and to increase channel depth.

Available now, the software cores' pricing is dependent on various factors including the actual implementation, licensing fees and volume of the order. **Equater Technologies, Inc. , 1300 White Oaks Rd., Campbell, CA 95008; (408) 369-5200; Fax: (408) 369-5210; www.equator.com.**

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National Semiconductor Unveils Software Platform, Reference Schematic

National Semiconductor Corp. recently announced availability of a software development platform and reference schematic for WebPAD designs to be used in conjunction with its Geode SC3200 MPU.

Geode SC3200 features a 32-bit × 86 compatible processor, a TFT video processor, core logic and a super I/O block along with a CCIR-656 video input port to enable media streaming acceleration. The SoC allows PAD designers to reduce component count and the number of traces on the main system board in addition to lowering overall system power requirements.

The company also has designed a software development platform for Geode SC3200 based WebPAD designs as well as Geode SC2200 thin client designs. Designated Geode SP4SC30, it enables users to develop code for various hardware options in a stable and highly observable system. The board eases software development through its inclusion of IDE, CRT and PCI bus interfaces and by providing access points for developing and debugging software. It also enables customers to evaluation the array of operating systems supporting the SC3200: BelA, Linux, NTe, QNX Neutrino and WinCE 3.0. Available now, the SP4SC30's, with power management and wireless communication options, price starts at $1520 each.

National also released reference schematics for the SC3200. Available free from National's web site, these schematics are designed for development of full-featured, small form-factor personal access devices taking advantage of the PAD-specific features of the SC3200. Next month, CoCom Group releases several versions of these reference schematics in a fully functioning wireless WebPAD appliance with multi-media functionality. These include upgrade kits, but the actual reference schematics price is about $1500. CoCom ordering information is available at www.cocomgroup.com. **National Semiconductorwww.national.com.**

Write in 5230 or **www.ecnmag.com/info**

Palmchip Introduces MP3 SoC hardware Platform

Palmchip Corp. recently announced an MP3 SoC hardware platform. The platform features a configurable 32-bit RISC/DSP processor from ARC Cores, Inc. The two companies will jointly market the platform.

Designated PALM-FC-6710, the SoC hardware platform supports multi-media card devices, audio input and output sources and serial interfaces, including UART and USB. It includes DMA support for peripheral interfaces to buffer memory for reduced firmware requirements and power management for low-power operation.

The platform integrates ARC's RISC processor with 8 KB of CPU SRAM. Its memory controller supports up to 4 MB of external memory or memory mapped ports. A second UART offers a debug communication port. The SoC development time can be reduced up to four months. By using a configurable processor, designers can easily comply with changing requirements. The platforms also contain the peripheral functions necessary for a basic SoC design, including system and watchdog timers, programmable I/O and interrupt and memory controllers.

Available now, the MP3 SoC hardware platform's price depends on various customer requirements including order volume. **Palmchip Corp., 2595 Junction Ave., Second Floor, San Jose, CA 95134; (408) 952-2000; Fax: (408) 570-0910; www.palmchip.com.**

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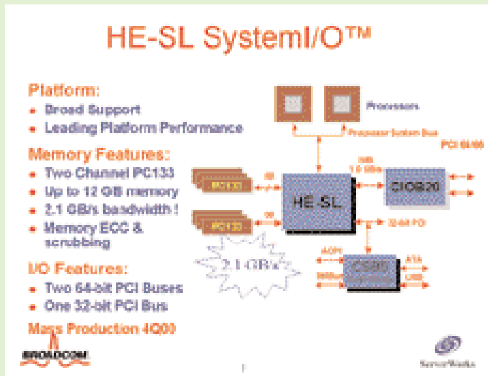
ServerWorks Offers System I/O Logic

ServerWorks, a subsidiary of Broadcom, recently announced its new system I/O logic, designated HE Super Lite or HE-SL or ServerSet III HE-SL. It was designed to meet the needs for dense Internet network servers that require maximum bandwidth in a small footprint.

HE-SL integrates support for two independent 64-bit, 66 MHz and one 32-bit, 33 MHz PCI buses to serve soaring data rates and dense, rack-mounted packaging in leading edge network systems. It supports Intel's Pentium III and Pentium III Zeon processors achieving speeds to 1 GHz in single and dual processor configurations. It is designed to support Intel's future 0.13-micron Pentium III CPUs when they are available.

The system I/O's memory controller offers memory bandwidth of 2.1 GBps through dual channels of PC133 registered DIMMs using a two-way interleaved operation. The design can accommodate up to six registered DIMMs with capabilities of 64 MB through 2 GB or up to 12 GB. It also incorporates an ECC algorithm that corrects and detect single-bit and multi-bit errors. Other features include a memory scrubbing capability that detects errors before these can impact system operation. Previous ServerWorks' systems required seven core logic functions on the system board, HE-SL reduces that to three for increased board space.



ServerWorks HE-SL system I/O integrates support for two independent 64-bit, 66 MHz and one 32-bit 33 MHz PCI buses to serve increasingly high data rates and dense, rack mounted packaging in leading edge network systems.

The HE-SL includes a North bridge (NB6576), a south bridge (SB7440) and a PCI bus bridge (NB6555). The three-chip set is priced at $80 each in 25,000-piece quantities. **ServerWorks, 2251 Lauson Lane, Santa Clara, CA 95054; (408) 492-1915; www.serverworks.com.**

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AMD Flash Offers New Architecture

Advanced Micro Devices, Inc. recently announced a new flash memory developed with its new flex bank architecture. The 64-Mb flash is dubbed Am29DL640. The new architecture permits designers to extend simultaneous read/write to multi-band devices. It is designed for various applications including advanced automotive, portable, handheld and cellular devices in addition to telecommunications, networking and set-top boxes.

Simultaneous read/write technology lets users read code from one bank and execute a program or erase a function in another bank with no switching latency. Most bank architectures limit designers to two fixed memory banks. Flex bank architecture extends the SRW technology to multiple bank devices allowing designers to change code and data memory partitions in real time enabling solutions tailored to end user needs.

AMD's new flash has four memory banks; two 8 Mb and two 24 Mb. It features the company's zero power operation permitting the device to put itself into sleep mode when inactive and limiting current consumption to 0.2 µA. AMD supports the Am29DL640 with its data management software. The software takes advantage of the flash by storing system software in one bank and storing data in the other.

Housed in TSOPs and FBGAs, the new flash contains SecSi™, the company's secured silicon sector to protect against unauthorized system cloning. Available now in either a TSOP or FBGA, the Am29DL640 is priced at $26.15 each in quantities of 10,000 pieces. **AMD One AMD Place, P.O. Box 3453, Sunnyvale, CA 94088; (800) 538-8450; www.amd.com.**

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National Semiconductor Debuts LVDS Family, LDO Regulator

National Semiconductor Corp. recently introduced a new series of high speed LVDS data transmission products and a 500 mA LDO voltage regulator.

The LVDS chips offer performance advantages, more integration, compact package design and on-chip test capability for telecommunications and datacom systems, mobile phone base stations and Internet access appliances. One device is the first in a series of IEEE 1149.1 JTAG compliant LVVDS default devices containing boundary scan test access. The standard defines a four-wire digital interface to a standard test access port on each compliant device.

National also has introduced the DS92LV16, a single-chip 16:1 LBDS serializer and a 1:16 deserializer. The 1.28 Gbps design can be used to construct 16 bi-directional point-to-point links across two pairs between two DS9LV16s. The chip also offers a flexible clocking scheme permitting variable input rates from 30 MHz to 80 MHz with a 5 percent clocking disparity between chips. It contains built-in local and line loopback modes to ease segregation of pre-specified parts of the system by the repetition of signals back to the board or back to the cable or backplane. It has a single passive termination resistor and is designed for VoIP, access multiplexers and routers. It also can provide a serial link across cable or backplanes in base station and Ethernet equipment and in line card-to-switch interfaces and proprietary video/imaging systems.

Available now, the DS92LV16 is housed in an 80-lead LQFP and is priced at $14 each in quantities of 1,000 pieces.

The DS92LV1260 offers six 1:10 LVDS deserializers in a single package. A seventh input channel, added as an alternate path to each deserializer input, offers built-in redundancy. This chip's design tracks base station distribution of digitized data from six ADC blocks corresponding to the six antenna sectors to multiple baseband processing boards. It also can be used in systems relating to Internet information appliance infrastructures. The next generation will include a stepped up clock rate to 66 MHz in addition to JTAG and BIST. It is priced at $39.80 each in 1,000 piece quantities.

National has designated DS90LV110 as a 1:10 data distributor. It allows designers to extend high speed, low power LVDS bus technology. It broadcasts 10 LVDS outputs from a single LVDS input permitting designers to proliferate LVDS data rates in backplane designs for cellular base stations. It also can be used in distributing a high-speed clock signal throughout large systems. The chip's receiver accepts LVPECL input signals or PECL levels with attenuation networks. It dissipates 800 mW so it doesn't need massive cooling or power supply needs. It also is available now, housed in a 28-pin SSOP and priced at $6 each in quantities of 1,000 units.

The CLC001 is a serial digital cable driver that receives LVDS signals and outputs PECL signals over 75 Ω cables used with digital routers and distribution amplifiers intermediate intra- or inter building communication. It features adjustable output voltage levels from 800 mV p-p to 1,000 mV p-p through a single external resistor. It also can move high-speed data across standard 75 Ω impedance cables with professional VCRs and digital cameras. It operates at speeds up to 622 Mbps. Housed in an 8-pin SOIC, the CLC001 is priced at $2 each in 1,000-piece quantities.

National's new DS90LV001 is an LVDS buffer housed in an 8-pin LLP. Inputs and outputs are LVDS compliant. Its shorter stubs increase data rates and improve performance. Its wide input dynamic range allows acceptance of LVPECL signals. It operates at up to 800 Mbps, comes in either an LLP or SOIC and is priced at $0.80 each in quantities of 1,000 pieces.

National also developed the DS92LV040, a four-channel bus LVDS transceiver. It contains two pairs of bus LVDS drivers and receivers with separate control signals to provide individual clock and data pair distribution through backplanes or cable interconnects. It buses 155 Mbps clock or data signals in telecommunications systems. Housed in a 44-lead LLP, it is priced at $4.50 each in 1,0000 piece quantities.

The 500 mA LDO voltage regulator is designed for medium level current uses in PC and portable systems that need low noise voltage conversion. Designated LP2989, the chip offers a 78 C/W power dissipation measured by theta j-a. It is designed for DSPs, graphics cards, PC and cellular phone audio subsystems, low voltage memories and bus terminals among others.

It offers a guaranteed 500 mA continuous output current along with an very low dropout voltage of 310 mV at a 500 mA load and 150 mV at a 200 mA load, typically. Its output noise measures 18 µV with a typical ground pin current of 1 mA at 200 mA loads and 3 mA at 500 mA loads. The chip also features a logic-controlled on/off. Its quiescent current drops to less than 1 µA in the shutdown mode. Voltage options range from 1.5 V to 5 V with the precise output voltage of plus or minus 1.5 per cent at room temperature over load and line variations.

Other features include an error flag that dips when the output drops 5 per cent below normal. The chip can be used in wake-on-LAN applications and contains protection for over-temperature and over current conditions. Its output capacitor requirements are 4.7 µF.

Available now, the LP2989 is housed in an LLP, an 8-pin SOIC or mini-SOIC surface mount packages. Prices range from $0.91 to $0.99 each, depending on the configuration, in quantities of 1,000 pieces. **National Semiconductor, 2900 Semiconductor Dr., Box 58090, Santa Clara, CA 95052; (800) 272-9959; www.national.com.**

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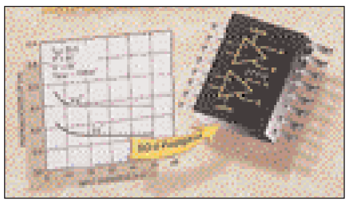
Semiconductor Monthly

Discrete Semiconductors

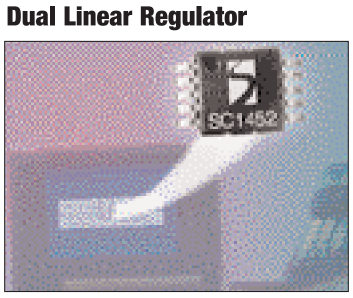
Railo-to-Rail Input/Output Comparator



The LT1712 Comparator features 4.5 ns propagation delay, a rail-to-rail input voltage range, complementary rail-to-rail CMOS/TTL compatible outputs, with internal data latches and an operating supply range of 2.4 V to 12 V. Its 2.4 V operation combined with its rail-to-rail inputs and CMOS compatible outputs makes it ideal for use in a wide variety of low voltage digital applications such as clock recovery, or as a line receiver in broadband communications systems. The LT1712's 12 V operational capability, high speed and small SSOP package makes it equally well suited for use in automatic test equipment and the dual internal latches are ideal for data sampling applications. Key specifications include: 4.5 ns at 20 mV Overdrive, 5 ns at 5 mV Overdrive; input common mode extends beyond supplies; specified at single 2.7 V/5 V and ±5 V supplies; and dual output latch capability. The product is available in a 16-pin SSOP package that occupies the same footprint as a conventional SO-8 package. Specified for operation over the industrial or the commercial temperature ranges, pricing starts at $3.35 each in 1,000-piece quantities. **Linear Technology Corp., 1630 McCarthy Blvd., Milpitas, CA 95035; (408) 432-1900; Fax: (408) 434-6441; www.linear-tech.com.**

Write in 5235 or **www.ecnmag.com/info**

Dual Linear Regulator



The SC1452 150mA dual linear regulator features small packaging, low quiescent current and low dropout voltage necessary for portable, battery-powered products. It is designed specifically to fully support a single-cell Li-Ion battery or four-cell NiMH or Alkaline applications. The device contains two independently enabled, ultra low dropoutvoltage regulators (ULDOs). It operates from an input voltage range of 2.5 V to 6.5 V with a wide variety of output voltage options available as standard offerings. Each regulator has an associated active-low reset signal that is asserted when the voltage output declines below the preset threshold. These reset signals are used to signal microprocessors, Digital Signal Processors (DSP) or other components that require notification when their proper input voltage is not available. Once the output recovers, the reset continues to be asserted (delayed) for a predetermined time (50ms for reset A and 150ms for reset B) to assure that the regulator is operating **Semtech Corp., 652 Mitchell Rd., Newbury Park, CA 91320; (805) 498-2111; info@semtech.com; www.semtech.com.**

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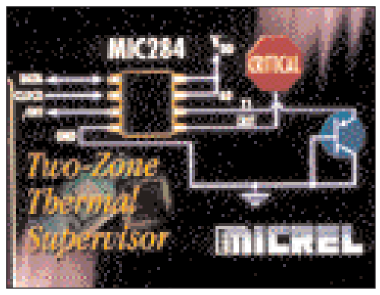
Thermal Supervisor IC



Micrel Semiconductor introduces a two-zone thermal supervisor IC for thermal management that uses embedded thermal diodes . The MIC284 is part of the family of system and thermal products, which is available in 8-pin MSOP and SOIC packaging. This product is capable of local and remote temperature measurement, I²C/SMBus compatibility, has a programmable interrupt output, an over-temperature alarm output and an address selection input. **Micrel Semiconductor, 1849 Fortune Dr., San Jose, CA 95131; (408) 944-0800; (408) 944-0970; www.micrel.com.**

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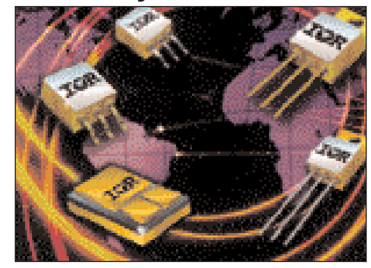
Hermetically Sealed Power MOSFETs



International Rectifier introduces their family of hermetically sealed high-reliability HEXFET® power MOSFETs for "step down" DC/DC power conversion systems in satellite launch vehicles, missiles, down-hole oil drilling and undersea broad band communications equipment. The product line includes 30 V, 55 V, 100 V and 150 V devices. The 30 V devices are targeted for 1.8 V and lower output DC/DC converters, the 55 V devices are used in 3.3 V and 5 V DC/DC converters, and the 100 V and 150 V products are used for 28 V to 100 V subsystems. **International Rectifier, 233 Kansas St., El Segundo, CA 90245; (310) 252-7105; www.irf.com.**

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